

```
In [141]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
%matplotlib inline
```

```
In [ ]: from sklearn.datasets import load_boston
boston = load_boston()
```

```
In [143]: data = pd.DataFrame(boston.data)
```

```
In [144]: data.columns = boston.feature_names
data.head()
```

```
Out[144]:
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	B
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1.0	296.0	15.3	396.90
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2.0	242.0	17.8	396.90
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2.0	242.0	17.8	392.83
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3.0	222.0	18.7	394.63
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3.0	222.0	18.7	396.90

```
In [145]: data['PRICE'] = boston.target
```

```
In [146]: data
```

```
Out[146]:
```

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	PRICE
0	0.00632	18.0	2.31	0.0	0.538	6.575	65.2	4.0900	1.0	296.0	15.3	396.90
1	0.02731	0.0	7.07	0.0	0.469	6.421	78.9	4.9671	2.0	242.0	17.8	396.90
2	0.02729	0.0	7.07	0.0	0.469	7.185	61.1	4.9671	2.0	242.0	17.8	392.83
3	0.03237	0.0	2.18	0.0	0.458	6.998	45.8	6.0622	3.0	222.0	18.7	394.63
4	0.06905	0.0	2.18	0.0	0.458	7.147	54.2	6.0622	3.0	222.0	18.7	396.90
...	...	...	...	...	...	...	...	...	...	...	...	...
501	0.06263	0.0	11.93	0.0	0.573	6.593	69.1	2.4786	1.0	273.0	21.0	391.90
502	0.04527	0.0	11.93	0.0	0.573	6.120	76.7	2.2875	1.0	273.0	21.0	396.90
503	0.06076	0.0	11.93	0.0	0.573	6.976	91.0	2.1675	1.0	273.0	21.0	396.90
504	0.10959	0.0	11.93	0.0	0.573	6.794	89.3	2.3889	1.0	273.0	21.0	393.40
505	0.04741	0.0	11.93	0.0	0.573	6.030	80.8	2.5050	1.0	273.0	21.0	396.90

506 rows × 14 columns

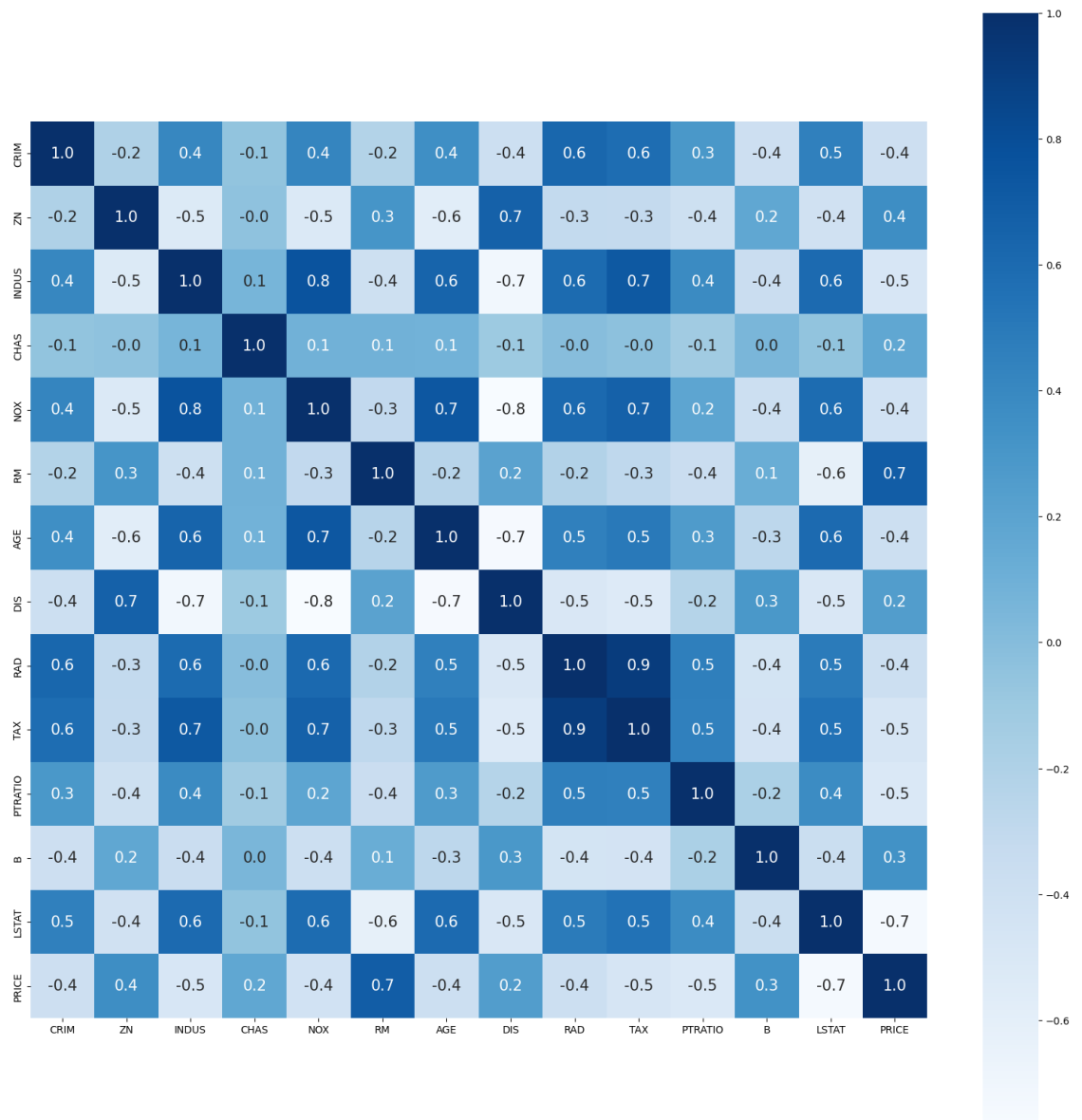
```
In [147]: data.isnull().sum()
```

```
Out[147]: CRIM      0
          ZN        0
          INDUS     0
          CHAS      0
          NOX       0
          RM        0
          AGE       0
          DIS       0
          RAD       0
          TAX       0
          PTRATIO   0
          B         0
          LSTAT     0
          PRICE     0
          dtype: int64
```

```
In [148]: corr = data.corr()
          corr.shape
```

```
Out[148]: (14, 14)
```

```
In [149]: plt.figure(figsize=(20,20))
sns.heatmap(corr, cbar=True, square=True, fm='.1f', annot=True, ann
plt.show())
```



```
In [150]: x = data.drop(['PRICE'], axis = 1)
y = data['PRICE']
```

```
In [151]: from sklearn.model_selection import train_test_split
xtrain, xtest, ytrain, ytest = train_test_split(x, y, test_size = 0.2,
```

```
In [152]: import sklearn
from sklearn.linear_model import LinearRegression
lm = LinearRegression()
# Train the model using the training sets
model=lm.fit(xtrain,ytrain)
```

In [153]: xtrain

Out[153]:

	CRIM	ZN	INDUS	CHAS	NOX	RM	AGE	DIS	RAD	TAX	PTRATIO	
220	0.35809	0.0	6.20	1.0	0.507	6.951	88.5	2.8617	8.0	307.0	17.4	391.1
71	0.15876	0.0	10.81	0.0	0.413	5.961	17.5	5.2873	4.0	305.0	19.2	376.9
240	0.11329	30.0	4.93	0.0	0.428	6.897	54.3	6.3361	6.0	300.0	16.6	391.1
6	0.08829	12.5	7.87	0.0	0.524	6.012	66.6	5.5605	5.0	311.0	15.2	395.0
417	25.94060	0.0	18.10	0.0	0.679	5.304	89.1	1.6475	24.0	666.0	20.2	127.1
...	...	...	...	...	...	...	...	...	...	...	...	...
323	0.28392	0.0	7.38	0.0	0.493	5.708	74.3	4.7211	5.0	287.0	19.6	391.1
192	0.08664	45.0	3.44	0.0	0.437	7.178	26.3	6.4798	5.0	398.0	15.2	390.4
117	0.15098	0.0	10.01	0.0	0.547	6.021	82.6	2.7474	6.0	432.0	17.8	394.1
47	0.22927	0.0	6.91	0.0	0.448	6.030	85.5	5.6894	3.0	233.0	17.9	392.1
172	0.13914	0.0	4.05	0.0	0.510	5.572	88.5	2.5961	5.0	296.0	16.6	396.1

404 rows × 13 columns

In [154]: ytrain_pred=lm.predict(xtrain)
ytest_pred=lm.predict(xtest)

In [155]: testdata=[[0.00632,18.0,2.31,0.0,0.538,6.575,65.2,4.0900,1.0,296.0,1

In [156]: test_pred = lm.predict(testdata)
test_pred

C:\ProgramData\Anaconda3\lib\site-packages\sklearn\base.py:450: Use
rWarning: X does not have valid feature names, but LinearRegression
was fitted with feature names
warnings.warn(

Out[156]: array([30.49949836])

```
In [157]: df1=pd.DataFrame(ytrain_pred,ytrain)
df2=pd.DataFrame(ytest_pred,ytest)
df1
```

```
Out[157]: 0
```

PRICE

26.7 32.556927

21.7 21.927095

22.0 27.543826

22.9 23.603188

10.4 6.571910

...

18.5 19.494951

36.4 33.326364

19.2 23.796208

16.6 18.458353

23.1 23.249181

404 rows × 1 columns

```
In [158]: from sklearn.metrics import mean_squared_error, r2_score
mse = mean_squared_error(ytest, ytest_pred)
print('MSE on test data:',mse)
mse1 = mean_squared_error(ytrain_pred, ytrain)
print('MSE on training data:',mse1)
```

MSE on test data: 33.44897999767653

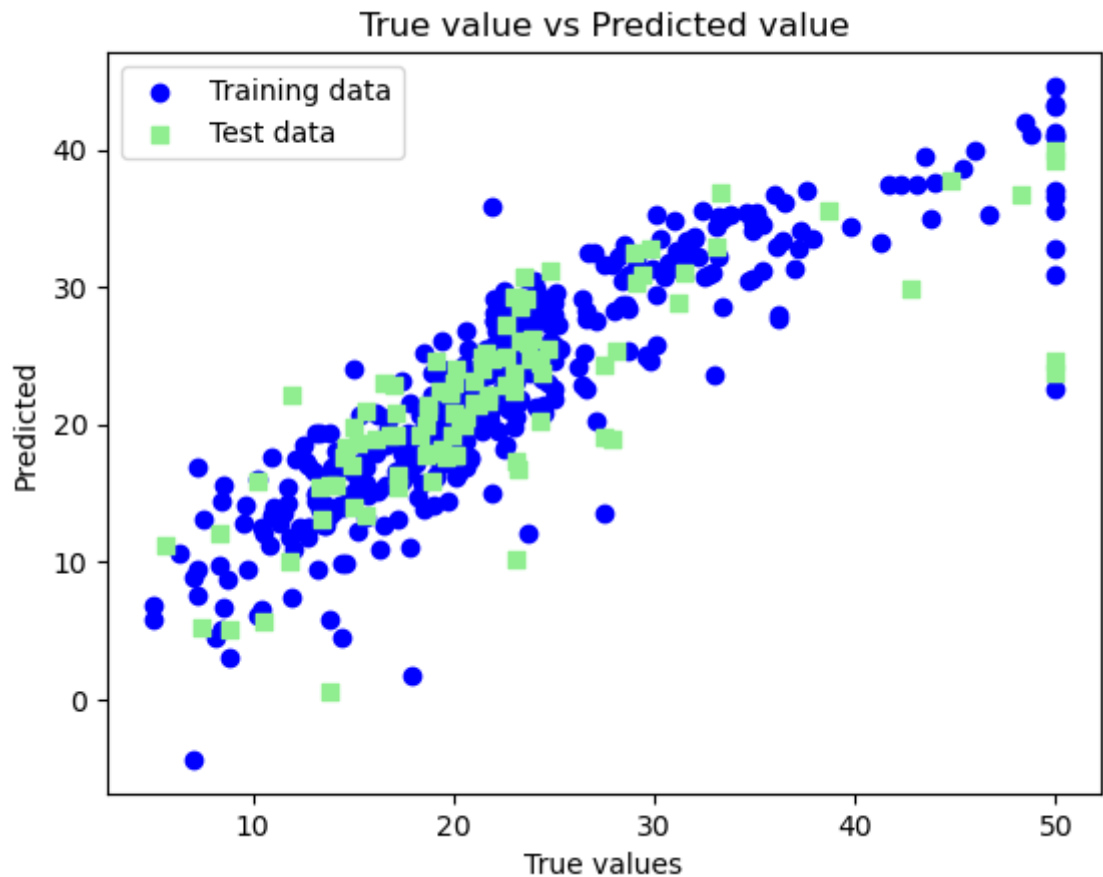
MSE on training data: 19.326470203585725

```
In [159]: #from sklearn.metrics import mean_squared_error
#def linear_metrics():
r2 = lm.score(xtest, ytest)
rmse = (np.sqrt(mean_squared_error(ytest, ytest_pred)))
print('r-squared: {}'.format(r2))
print('-----')
print('root mean squared error: {}'.format(rmse))
```

r-squared: 0.5892223849182507

root mean squared error: 5.783509315085135

```
In [160]: #plotting the linear regression model
plt.scatter(ytrain ,ytrain_pred,c='blue',marker='o',label='Training')
plt.scatter(ytest,ytest_pred ,c='lightgreen',marker='s',label='Test')
plt.xlabel('True values')
plt.ylabel('Predicted')
plt.title("True value vs Predicted value")
plt.legend(loc= 'upper left') #plt.hlines(y=0,xmin=0,xmax=50)
plt.plot()
plt.show()
```



```
In [161]: testdata=[ [0.00632,18.0,2.31,0.0,0.538,6.575,65.2,4.0900,1.0,296.0,1
```

```
In [162]: test_pred = lm.predict(testdata)
test_pred
```

C:\ProgramData\Anaconda3\lib\site-packages\sklearn\base.py:450: Use
rWarning: X does not have valid feature names, but LinearRegression
was fitted with feature names
warnings.warn(

```
Out[162]: array([30.49949836])
```

```
In [ ]:
```